Low-Cost Digital Oscilloscope for Lissajous Visualization and Real-Time Signal Analysis

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Abstract

Oscilloscopes are essential tools in experimental physics but are often hard to access outside labs. This project demonstrates a low-cost, two-channel digital approximation of an oscilloscope, making real-time waveform visualization possible without specialized equipment. The system takes two simultaneous audio inputs and treats them as independent measurement channels. From each channel, the dominant frequency component is extracted using Fourier analysis, and an approximate sinusoidal representation is generated. In this process, the remaining components are suppressed to obtain a simplified signal. The inputs from the two audio channels are then applied to perpendicular axes, and their combination produces Lissajous figures. The shape of these figures carries information about the phase difference and frequency ratio between the two inputs, and they update in real time as the signals change, which gives an immediate and visual way to study the relationship between two signals. This project shows that several principles underlying standard oscilloscope functions can be carried out with simple, low-cost hardware. The aim is not to replace laboratory instruments, but to demonstrate that the underlying physics is accessible and observable to all.

Mathematical Basis of Lissajous Figures & Application

Parametric Representation

1. Signal Definition
The superposition of two orthogonal harmonic oscillations is given by:

$$x(t) = A\sin(\omega_1 t), \quad y(t) = B\sin(\omega_2 t + \delta)$$

Where δ represents the relative phase shift between the two channels.

2. Frequency Ratio & Geometry
Closed, stationary trajectories (Lissajous figures) occur only when the frequency ratio is a rational number:

$$\frac{\omega_1}{\omega_2} = \frac{n}{m} \quad (n, m \in \mathbb{Z})$$

Observation Rule: The ratio of the number of horizontal tangents to vertical tangents equals the frequency ratio $\frac{n}{m}$.

3. The General Equation ($\omega_1 = \omega_2$)
For equal frequencies, the parametric equations combine into the general equation of an ellipse:

$$\left(\frac{x}{A}\right)^2 + \left(\frac{y}{B}\right)^2 - 2\frac{xy}{AB}\cos\delta = \sin^2\delta$$

Special Cases for Phase Shift δ:

- $\delta = 0, \pi$: A straight line ($y = \pm\frac{B}{A}x$).
- $\delta = \pi/2, 3\pi/2$: A principal ellipse (becomes a circle if $A = B$).

Digital Signal Reconstruction

I. Discrete Sampling
The sound card converts continuous voltage into a sequence of digital values at a sampling rate f_s (typically 44.1 kHz). The time between samples is:

$$\Delta t = \frac{1}{f_s}$$

II. Fundamental Frequency Detection
Real-world audio contains noise and harmonics. The system identifies the strongest frequency component f by analyzing the signal energy. We approximate the "true" wave as:

$$S_{ideal}(t) = A\sin(2\pi f t + \phi)$$

This allows the visualization of a stable Lissajous figure even if the raw input is slightly distorted.

III. Visual Mapping
The digital oscilloscope plots the reconstructed pairs (x_n, y_n) in real-time. By comparing the raw input S_{raw} and the reconstructed model S_{ideal}, we can calculate the **Residual Signal (Noise)**:

$$\text{Noise Component} = S_{raw}(t) - S_{ideal}(t)$$

Experimental Measurement Logic

I. The Method of Tangents
The frequency ratio of the two inputs can be determined visually by counting the points of tangency at the boundaries of the figure:

$$\frac{f_y}{f_x} = \frac{\text{Number of Horizontal Tangents}}{\text{Number of Vertical Tangents}}$$

II. Phase Extraction from Intercepts
For a 1:1 frequency ratio, the phase difference δ is calculated using the ratio of the y-intercept y_0 to the maximum vertical amplitude B:

$$\delta = \arcsin\left(\frac{y_0}{B}\right)$$

This allows the digital oscilloscope to function as a precise phase-meter for real-time signal analysis.

Software Logic & Signal Processing

I. Automatic Scaling (Normalization)
To keep the plot visible regardless of the mobile's volume, the code divides the raw signal by its own maximum value (M):

$$\text{Output} = \frac{\text{Input Signal}}{\text{Peak Amplitude (M)}}$$

This ensures the wave always stays within the display boundaries (-1 to $+1$).

II. Stability through Smoothing
Raw frequency detection often "jumps" due to background noise. To keep the drawing steady, the software uses a weighted average:

$$f_{stable} = (0.9 \times f_{previous}) + (0.1 \times f_{current})$$

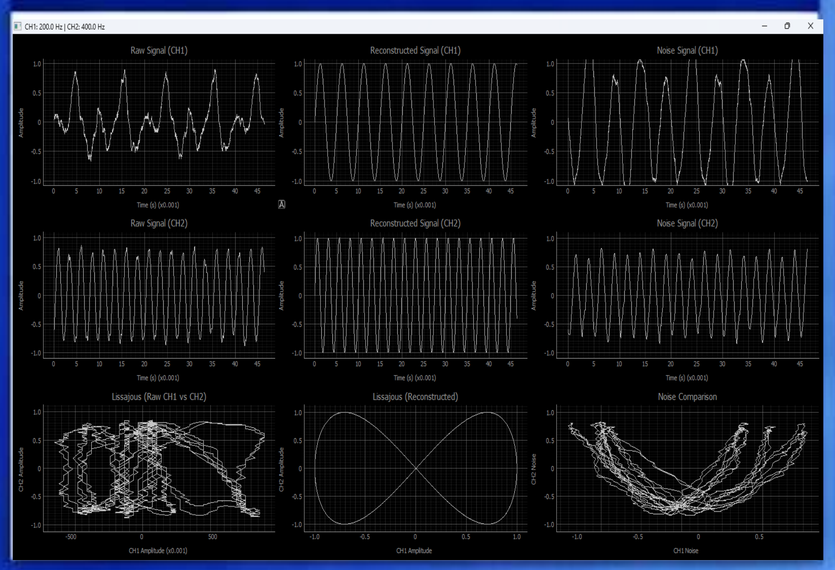
This "inertia" prevents the Lissajous figure from flickering, providing a smooth real-time visualization.

III. Identifying Experimental Noise
By creating a mathematically "perfect" sine wave and subtracting it from the raw recording, we isolate the background interference:

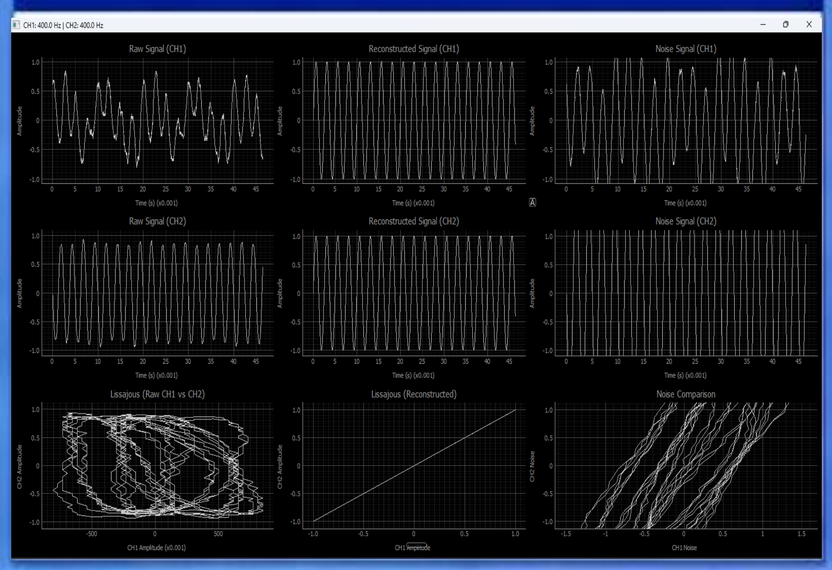
$$\text{Noise} = \text{Recorded Signal} - \text{Ideal Sine Wave}$$

This allows us to visualize exactly how much "hum" or distortion the microphone is picking up from the environment.

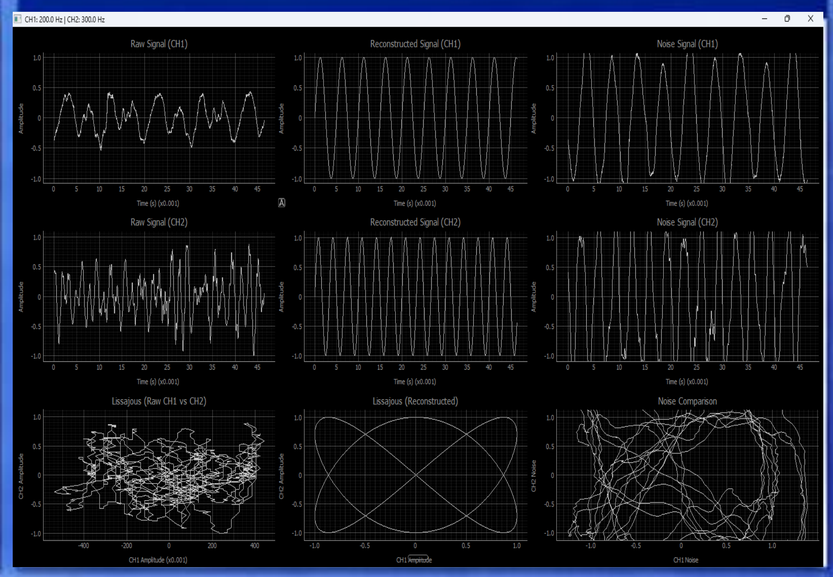
Lissajous (1:2 Frequency Ratio)



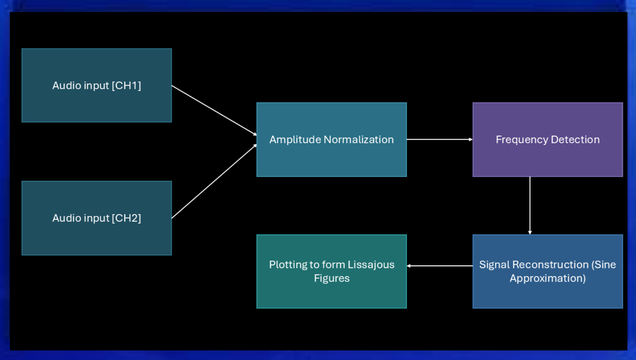
Lissajous (1:1 Frequency Ratio)



Lissajous (2:3 Frequency Ratio)



Flowchart



Observations

- Reconstructed Waveform closely matches steady sinusoidal inputs
- Lissajous patterns forms stable and distinct patterns for simple frequency ratios (e.g. 1:1 or 1:2 or 2:3)

Key Insights

- Real signals are not perfectly sinusoidal due to noise and higher harmonics creeping in
- Using only the dominant frequency gives an approximation, not the full signal
- Lissajous patterns depend on frequency ratios and phase
- Residual signal (Noise component) shows how much the real signal varies from the ideal model

Conclusion

- A simple setup can demonstrate basic oscilloscope and signal analysis concepts
- Real time visualization helps understand the dependence of Lissajous figures on the frequency directly.
- The project connects theoretical components like waves, frequency with actual measurements which can be altered physically.

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